

Chapter 2

Frequency Distributions and Graphs

A **frequency distribution** is the organization of raw data in table form, using classes and frequencies.

Categorical Frequency Distributions

The **categorical frequency distribution** is used for data that can be placed in specific categories, such as nominal- or ordinal-level data. For example, data such as political affiliation, religious affiliation, or major field of study would use categorical frequency distributions.

EXAMPLE 2-1 Distribution of Blood Types

Twenty-five army inductees were given a blood test to determine their blood type. The data set is

A	B	B	AB	O
O	O	B	AB	B
B	B	O	A	O
A	O	O	O	AB
AB	A	O	B	A

Construct a frequency distribution for the data.

SOLUTION

A Class	B Tally	C Frequency	D Percent
A	///	5	20
B	/// //	7	28
O	/// ////	9	36
AB	///	4	16
		<u>25</u>	<u>100%</u>

For the sample, more people have type O blood than any other type.

Grouped Frequency Distributions

When the range of the data is large, the data must be grouped into classes that are more than one unit in width, in what is called a **grouped frequency distribution**. For example,

EXAMPLE 2-2 Record High Temperatures

These data represent the record high temperatures in degrees Fahrenheit (°F) for each of the 50 states. Construct a grouped frequency distribution for the data, using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114

Source: *The World Almanac and Book of Facts*.

SOLUTION

The procedure for constructing a grouped frequency distribution for numerical data follows.

Determine the classes.

Find the highest value and lowest value: $H = 134$ and $L = 100$.

Find the range: $R = \text{highest value} - \text{lowest value} = H - L$, so

$$R = 134 - 100 = 34$$

Select the number of classes desired (usually between 5 and 20). In this case, 7 is arbitrarily chosen.

Find the class width by dividing the range by the number of classes.

$$\text{Width} = \frac{R}{\text{number of classes}} = \frac{34}{7} = 4.9$$

Round the answer up to the nearest whole number if there is a remainder: $4.9 \approx 5$.

The completed frequency distribution is

Class limits	Class boundaries	Tally	Frequency
100–104	99.5–104.5	//	2
105–109	104.5–109.5		8
110–114	109.5–114.5		18
115–119	114.5–119.5		13
120–124	119.5–124.5		7
125–129	124.5–129.5	/	1
130–134	129.5–134.5	/	1
			Total 50

↓
 $2 + 8 + 18 = 28$

The frequency distribution shows that the class 109.5–114.5 contains the largest number of temperatures (18) followed by the class 114.5–119.5 with 13 temperatures. Hence, most of the temperatures (31) fall between 110 and 119°F.

The cumulative frequency distribution for the data in this example is as follows:

	Cumulative frequency
Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50

EXAMPLE 2-3 MPG for SUVs

The data shown here represent the number of miles per gallon (mpg) that 30 selected four-wheel-drive sport utility vehicles obtained in city driving. Construct a frequency distribution, and analyze the distribution.

12	17	<u>12</u>	<u>14</u>	<u>16</u>	<u>18</u>
16	18	12	16	<u>17</u>	<u>15</u>
15	16	12	15	16	16
12	14	15	12	15	15
19	13	16	18	16	14

Source: Model Year Fuel Economy Guide, United States Environmental Protection Agency.

SOLUTION

Step 1 Determine the classes. Since the range of the data set is small ($19 - 12 = 7$), classes consisting of a single data value can be used. They are 12, 13, 14, 15, 16, 17, 18, 19.

Note: If the data are continuous, class boundaries can be used. Subtract 0.5 from each class value to get the lower class boundary, and add 0.5 to each class value to get the upper class boundary.

Step 2 Tally the data.

Step 3 Find the numerical frequencies from the tallies, and find the cumulative frequencies.

The completed ungrouped frequency distribution is

Class limits	Class boundaries	Tally	Frequency
<u>12</u>	<u>11.5–12.5</u>	<u> /</u>	<u>6</u>
<u>13</u>	<u>12.5–13.5</u>	<u>/</u>	<u>1</u>
<u>14</u>	<u>13.5–14.5</u>	<u> </u>	<u>3</u>
<u>15</u>	<u>14.5–15.5</u>	<u> /</u>	<u>6</u>
<u>16</u>	<u>15.5–16.5</u>	<u> / </u>	<u>8</u>
<u>17</u>	<u>16.5–17.5</u>	<u> </u>	<u>2</u>
<u>18</u>	<u>17.5–18.5</u>	<u> </u>	<u>3</u>
<u>19</u>	<u>18.5–19.5</u>	<u>/</u>	<u>1</u>

In this case, almost one-half (14) of the vehicles get 15 or 16 miles per gallon. The cumulative frequencies are

	Cumulative frequency
Less than 11.5	0
Less than 12.5	6
Less than 13.5	7
Less than 14.5	10
Less than 15.5	16
Less than 16.5	24
Less than 17.5	26
Less than 18.5	29
Less than 19.5	30

Exercises 2-1

- 13. Favorite Sport** Fifty people were asked to identify their favorite sport. The responses were F = professional football, C = college football, B = baseball, and A = auto racing. Construct a categorical frequency distribution for the data. Which class has the most data values, and which class has the fewest data values? (Based on information from a Harris Interactive survey.)

F	C	F	A	B
C	F	F	B	C
A	B	C	C	F
F	A	F	C	A
C	C	F	F	C
F	F	B	C	F
B	A	F	B	F
F	C	B	A	C
B	F	C	F	A
B	B	F	C	F

- 17. Energy Used by Plasma TVs** Forty plasma TVs were tested, and the number of watts per hour that they used while running for 4 hours per day were recorded. Construct a frequency distribution for the data. Use 6 classes.

480	503	514	472	465
473	475	500	502	506
492	477	483	475	481
512	473	488	485	503
508	484	486	499	512
485	468	507	501	470
497	479	492	468	492
466	472	483	511	481